

★ POINTS VS CHANGE + POPULATION % ★

Quant | Ch03 Averages + Profit + Interest | Topic 1 of 12

► Meaning in one line

Points (subtract.

► Key points

Point	Example
points = new% – old%.	40% → 46% = +6 points (NOT +15%!).
people/income/spend + growth.	pop 50k +12%: $50k \times 1.12 = 56k$.

★ IMPORTANT FOR EXAM

Averages

⚠ TRAP - read twice

Points-as-percent + '+6 points = +6%' + base-mix

✓ **Try yourself:** 45% → 54%? Pop 80k –15%? **Answer:** +9 points (+20%); 68k.

★ SIMPLE + MISSING AVERAGE ★

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► Meaning in one line

Sum/count , total , back-calculate.

► Key points

Point	Example
$\text{avg} = \text{sum}/n$.	12,18,20,25,25 $\rightarrow 100/5 = 20$.
$\text{miss} = \text{avg} \times n - \text{known}$.	avg 5 nums 20, known sum 82 $\rightarrow$ miss 18.

★ IMPORTANT FOR EXAM

Combined + weighted + speed

⚠ TRAP - read twice

Divide-count-wrong + 'average = middle' + miss-guess

✓ **Try yourself:** Avg 8 nums = 15, 7 sum 96. Miss? **Answer:** Total 120 – 96 = 24.

★ COMBINED + WEIGHTED AVERAGE ★

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► Meaning in one line

Total/total-n , weight×value/sum-weights , mix.

► Key points

Point	Example
$(n_1a_1+n_2a_2)/(n_1+n_2)$.	$30@40 + 20@55 = (1200+1100)/50 = 46$.
$\text{sum}(w \times x)/\text{sum}(w)$.	$70 \times 3 + 90 \times 1 / 4 = 300/4 = 75$.

★ IMPORTANT FOR EXAM

Speed + CP/SP/MP

⚠ TRAP - read twice

Simple-mean-of-means + 'average averages' + weights-ignored

✓ **Try yourself:** $40@60 + 60@90$ combined? 2:3 weights 50/80? **Answer:** $7800/100 = 78$; $(100+240)/5 = 68$.

★ AVERAGE SPEED + CP/SP/MP ★

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► Meaning in one line

Total-dist/total-time (harmonic).

► Key points

Point	Example
$2xy/(x+y)$ equal-dist.	$40 + 60 \rightarrow 2 \times 40 \times 60 / 100 = 48$ (not 50!).
CP + MP + SP + list.	CP500 MP800 SP680: disc = $120/800 = 15\%$.

★ IMPORTANT FOR EXAM

P&L; + discount + successive

⚠ TRAP - read twice

Arithmetic-mean-speed + 'CP = SP' + MP/SP-swap

✓ **Try yourself:** 60 + 90 equal-dist? CP300 SP375 profit%? **Answer:** $2 \times 60 \times 90 / 150 = 72$; $75/300 = 25\%$.

★ PROFIT/LOSS + DISCOUNT ★

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► Meaning in one line

P&L; on CP.

► Key points

Point	Example
$P\% = (SP - CP) / CP \times 100.$	CP200 SP250 = +25%; SP170 = -15%.
$D\% = (MP - SP) / MP \times 100.$	MP1000 SP850 = 15%.

★ IMPORTANT FOR EXAM

Successive + MP-profit + false-weight

⚠ TRAP - read twice

P&L-on-SP; + discount-on-CP + 'profit = SP' + base-mix

✓ **Try yourself:** CP400 SP460? MP2000 SP1700 disc%? **Answer:** +15%; 300/2000 = 15%.

★ SUCCESSIVE DISCOUNT + MP PROFIT ★

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► Meaning in one line

a, b, $ab/100$, net, chain, MP, CP, SP, margin.

► Key points

Point	Example
net = $a+b+ab/100$.	20+10: $20+10+2 = 32\%$? No: $-20 -10$: $-20-10+2 = -28\%$.
want 20% on CP600, MP800: SP = 720; disc = $80/800 = 10\%$.	-

★ IMPORTANT FOR EXAM

False-weight + SI + P/R/T

⚠ TRAP - read twice

Add-discounts + 'net = sum' + MP/CP-mix + margin-miss

✓ **Try yourself:** Net: $-15 -10$? CP500 MP650, want +30%? **Answer:** $-15-10+1.5 = -23.5\%$; SP650?
 $500 \times 1.3 = 650 \rightarrow 0\%$ disc.

★ FALSE WEIGHT + SIMPLE INTEREST ★

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► Meaning in one line

Claim/actual , gain% , cheat.

► Key points

Point	Example
$\text{gain\%} = (\text{claim} - \text{actual}) / \text{actual} \times 100.$	1000 vs 950: $50/950 = 5.26\%$.
$\text{SI} = \text{PRT}/100; A = P + \text{SI}.$	5000, 8%, 3y = 1200; A = 6200.

★ IMPORTANT FOR EXAM

Find-P/R/T + SI-words + compounding

⚠ TRAP - read twice

Gain-on-claim + 'SI = compound' + rate/time-mix

✓ **Try yourself:** 950 for 1000 gain%? SI: 8000, 5%, 4y? **Answer:** $50/950 = 5.26\%$; 1600.

★ FIND P/R/T + SI WORD PROBLEMS ★

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► Meaning in one line

$P = SI \times 100 / RT$, R, T, back.

► Key points

Point	Example
P/R/T from $SI \times 100 / RT$.	SI900, 6%, 3y $\rightarrow P = 900 \times 100 / 18 = 5000$.
double/mix/compare + stories.	double in 8y $\rightarrow R = 100 / 8 = 12.5\%$.

★ IMPORTANT FOR EXAM

Compounding (annual/half) + difference

⚠ TRAP - read twice

SI-for-P + rate/time-flip + 'double = $2 \times \text{rate}$ ' + no-back

✓ **Try yourself:** SI600, 5%, 4y $\rightarrow P$? Triple in 20y $\rightarrow R$? **Answer:** $P = 3000$; $R = 200 / 20 = 10\%$.

★ ANNUAL + HALF-YEARLY COMPOUNDING ★

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► Meaning in one line

$A = P(1, r/100)^n$, rate-half , time-double , effective.

► Key points

Point	Example
$A = P(1+r/100)^n$.	10000, 10%, 2y = $10000 \times 1.21 = 12100$; CI = 2100.
$r/2, 2n$.	10000, 10%, 1y half-yearly: $10000 \times (1.05)^2 = 11025$.

★ IMPORTANT FOR EXAM

SI-vs-CI + growth + partnership

⚠ TRAP - read twice

SI-formula + 'half = half-SI' + rate/time-unadjusted + no-power

✓ **Try yourself:** CI: 5000, 8%, 2y? Half-yearly 1y? **Answer:** $5000 \times 1.1664 = 5832 \rightarrow 832$; $5000 \times (1.04)^2 = 5408$.

★ SI-CI DIFFERENCE + GROWTH/DEPRECIATION ★

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► Meaning in one line

2-yr: $PR^2/100^2$.

► Key points

Point	Example
2-yr diff = $PR^2/100^2$.	10000, 10%: diff = 100.
$P(1 \pm r/100)^n$.	pop 80000 +5% 2y = $80000 \times 1.1025 = 88200$.

★ IMPORTANT FOR EXAM

Partnership (capital-time)

⚠ TRAP - read twice

CI-SI-guess + 'growth = SI' + sign-mix

✓ **Try yourself:** Diff: 4000, 5%, 2y? Decay 10% on 50000, 2y? **Answer:** $4000 \times 25/10000 = 10$; $50000 \times 0.81 = 40500$.

★ CAPITAL-TIME + JOIN/LEAVE PARTNERS ★

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► Meaning in one line

Share = $C \times T$, ratio , total.

► Key points

Point	Example
ratio $C_1T_1:C_2T_2$.	$5000 \times 12 : 8000 \times 6 = 60000:48000 = 5:4$; profit 9000 $\rightarrow$ 5000,4000.
join/leave + salary-first.	A 12mo, B 8mo + B salary 500, profit 6500: ratio first, then split rest.

★ IMPORTANT FOR EXAM

Weighted-average + alligation

⚠ TRAP - read twice

Capital-only + 'equal split' + time-ignored + salary-last

✓ **Try yourself:** $6000 \times 9 : 9000 \times 6$, profit 13000? **Answer:** $54000:54000 = 1:1 \rightarrow$ 6500 each.

★ SALARY + WEIGHTED AVERAGE (PARTNERS) ★

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► Meaning in one line

Salary/commission-first , rest-ratio.

► Key points

Point	Example
salary-first, split rest.	profit 10000, B salary 1000: rest 9000 in 2:1 → A6000, B1000+3000.
$\text{sum}(w \times x) / \text{sum}(w)$.	marks 3:2, 60/90: $(180+180)/5 = 72$.

★ IMPORTANT FOR EXAM

Alligation + replacement + dilution

⚠ TRAP - read twice

Split-then-salary + 'average averages' + weights-ignored

✓ **Try yourself:** Profit 13000, A salary 1000, ratio 1:1 rest? Weights 1:4, 40/100? **Answer:** Rest 12000 → $6000 + (1000+6000); (40+400)/5 = 88$.